Bidirectional Backpropagation Neural Machine Translation

Experiment v4: Untied Output Projections and Removal of Input Feeding

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1 Overview

This report documents Experiment v4 of the Bidirectional Backpropagation (B-BP) Neural Machine Translation project under the Adigun-Kosko framework. The core B-BP claim is that a single recurrent matrix $\mathbf{U} \in \mathbb{R}^{d \times d}$ can govern both forward translation and its algebraic inverse via weight transposition:

$$\text{Forward } (bbp\text{=False}): \quad h_t = \tanh(h_{t-1}\,\mathbf{U} + x_t + b) \tag{1}$$

$$\text{Backward } (bbp\text{=True}): \quad h_t = \tanh\Big(h_{t-1}\,\mathbf{U}^\top + x_t + b\Big) \tag{2}$$

If $\mathbf{U}$ is orthogonal, $\mathbf{U}^\top = \mathbf{U}^{-1}$ and the backward pass is the true algebraic inverse of the forward pass. This is the Adigun-Kosko inversion principle applied to sequence-to-sequence neural machine translation.

1.1 Experiment History

	v1	**v2**	**v3**	**v4**
Tokenisation	word-10k	BPE-8k	BPE-8k	BPE-8k
Attention	none	dot	dot	dot
Encoder layers	1	1	2	2
Input feeding	no	yes	yes	**no**
Output projection	tied	tied	tied	**untied**
Cycle warmup	no	no	yes	yes
Learning rate	3e-4	3e-4	1e-3	1e-3
EN→DE BLEU	5.37	≈ 0	0.01	**4.10**
DE→EN BLEU	3.47	≈ 0	0.00	**6.44**
Avg BLEU	4.42	≈ 0	0.01	**5.27**
Best val loss	17.08	14.18	12.61	**7.82**

2 Motivation: Problems Diagnosed in v3

Experiment v3 achieved monotonically decreasing validation loss (14.27 → 12.61 over 20 epochs) but produced near-zero BLEU throughout. After training, the decoder was probed step-by-step on the sentence "the house is on the left side of the street". The output was:

Step	**Token**	**Top-5 (token, prob)**
1	`die`	die 0.087, das 0.086, der 0.061, ist 0.029, in 0.029
2	`die`	die 0.061, . 0.043, der 0.034, das 0.028, ist 0.022
3	`die`	die 0.054, . 0.051, der 0.034, das 0.025, ist 0.020
4–10	`die`	*distribution identical to step 3*

Two distinct failure mechanisms were identified:

2.1 Failure 1: Decoder Fixed-Point Collapse via Input Feeding

v3 used input feeding (Luong et al., 2015): the decoder input at step t was $x_t = \text{Embed}(y_{t-1}) + \text{context}_{t-1}$. Once the decoder output "die" at step 1, the following chain locked the decoder into a fixed point:

$$h_2 = f(h_1,\ \text{Embed}(\texttt{die}) + \text{ctx}_1) \approx h_1 \tag{3}$$

$$\text{ctx}_2 \approx \text{ctx}_1 \quad \Rightarrow \quad x_3 \approx x_2 \quad \Rightarrow \quad h_3 \approx h_2 \tag{4}$$

Every subsequent step received nearly identical input, producing identical hidden states and identical output distributions. The maximum probability never exceeded 8.7%, confirming the distribution never sharpened.

2.2 Failure 2: Tied Embedding Lock

v3 used tied output weights: $\text{logit} = \mathbf{h} \cdot \mathbf{E}_{\text{tgt}}^{\top}$. The embedding matrix $\mathbf{E}_{\text{tgt}}$ was required to simultaneously satisfy three conflicting objectives:

1. Map token IDs to dense vectors for encoder and decoder input.
2. Be compatible with $\mathbf{U}^{\top}$ inversion (B-BP constraint).
3. Project decoder hidden states to sharp output distributions.

The gradient competition from these three objectives prevented the output distribution from sharpening. Argmax always selected the highest-frequency German article.

3 Architecture: v4

3.1 Fix 1: Untied Output Projections

Two dedicated output projection layers are added, one per language:

$$\mathbf{W}_{\text{out,en}} \in \mathbb{R}^{d \times V_{\text{en}}}, \quad \mathbf{W}_{\text{out,de}} \in \mathbb{R}^{d \times V_{\text{de}}} \tag{5}$$

The output logit is computed as:

$$\text{logit}_t = \mathbf{W}_{\text{out,tgt}}(h_t + \text{ctx_proj}(\text{context}_t)) \tag{6}$$

The embedding matrices $\mathbf{E}_{\text{en}}$ and $\mathbf{E}_{\text{de}}$ now serve only the input side. They are freed from output scoring duties and can focus entirely on building useful input representations.

3.2 Fix 2: Removal of Input Feeding

The decoder input at step t is simplified to embedding only:

$$x_t = \text{Embed}(y_{t-1}) \tag{7}$$

The attention context context_t still contributes at the output combination step — the attention benefit is fully preserved. Removing context from the input eliminates the fixed-point mechanism while retaining all information flow from the encoder to the output.

3.3 Components Unchanged from v3

The following are identical to v3:

- **B-BP RNN Cell**: $h_t = \tanh(h_{t-1}\mathbf{W} + x_t + b)$, where $\mathbf{W} = \mathbf{U}$ (forward) or $\mathbf{W} = \mathbf{U}^\top$ (backward). Single matrix $\mathbf{U}$ shared across all encoder layers and the decoder.
- **2-layer stacked encoder**: both layers share $\mathbf{U}$.
- **Scaled dot-product attention**: no learned scoring parameters.
- **BPE tokenization**: SentencePiece, 8,000 tokens per language.
- **Cycle warmup schedule**: $\lambda = 0.00$ (ep 1–5), $\lambda = 0.05$ (ep 6–10), $\lambda = 0.10$ (ep 11–20).
- **Learning rate**: 1×10^{-3}.

3.4 Four Inference Modes

Mode	Encoder W	Decoder W	Input Emb	Output Proj
EN→DE	$\mathbf{U}$	$\mathbf{U}$	$\mathbf{E}_{\text{en}}$ / $\mathbf{E}_{\text{de}}$	$\mathbf{W}_{\text{out,de}}$
DE→EN	$\mathbf{U}^\top$	$\mathbf{U}^\top$	$\mathbf{E}_{\text{de}}$ / $\mathbf{E}_{\text{en}}$	$\mathbf{W}_{\text{out,en}}$
EN cycle	$\mathbf{U}$	$\mathbf{U}^\top$	$\mathbf{E}_{\text{en}}$ / $\mathbf{E}_{\text{en}}$	$\mathbf{W}_{\text{out,en}}$
DE cycle	$\mathbf{U}^\top$	$\mathbf{U}$	$\mathbf{E}_{\text{de}}$ / $\mathbf{E}_{\text{de}}$	$\mathbf{W}_{\text{out,de}}$

3.5 Stacked Encoder Detail

$$h_t^{(1)} = \tanh\Big(h_{t-1}^{(1)}\mathbf{W} + \text{Embed}(x_t) + b_{\text{enc1}}\Big) \tag{8}$$

$$h_t^{(2)} = \tanh\Big(h_{t-1}^{(2)}\mathbf{W} + h_t^{(1)} + b_{\text{enc2}}\Big) \tag{9}$$

Attention attends over $\mathbf{H}^{(2)} = [h_1^{(2)}, \ldots, h_T^{(2)}]$. Decoder initialised from $h_T^{(2)}$. Both layers use the same $\mathbf{W}$ ($\mathbf{U}$ or $\mathbf{U}^\top$).

3.6 Training Objective

$$\mathcal{L} = \mathcal{L}$$

where $\mathcal{L}_{\text{cyc,EN}}$: encode with $\mathbf{U}$, decode with $\mathbf{U}^\top$, reconstruct EN; $\mathcal{L}_{\text{cyc,DE}}$: encode with $\mathbf{U}^\top$, decode with $\mathbf{U}$, reconstruct DE.

4 Model Parameters

Component	Parameters	Notes
`cell.U` (recurrent)	65,536	Shared across enc-L1, enc-L2, and decoder; used as $\mathbf{U}$ and $\mathbf{U}^\top$
`E_en` (input emb)	2,048,000	BPE 8k $\times$ 256, input only
`E_de` (input emb)	2,048,000	BPE 8k $\times$ 256, input only
`W_out_en`	2,048,000	Untied EN output projection (Fix 1)
`W_out_de`	2,048,000	Untied DE output projection (Fix 1)
`ctx_proj`	65,536	Attention context combiner
Biases (`b_enc1`, `b_enc2`, `b_dec`)	768	
Other	256	
Total	**8,324,096**	

Hyperparameters: `hid=256`, `batch=64`, `lr=1e-3`, `enc_layers=2`, `epochs=20`, `max_len=40`, `input_feeding=False`, `dropout=0.3`, `grad_clip=1.0`.

5 Experimental Setup

- **Dataset**: WMT14 EN-DE. 200,000 sentence pairs subsampled from 4,508,785 training pairs; filtered to max 40 BPE tokens per sentence.
- **Final split sizes**: Train 131,883 | Val 2,255 (newstest) | Test 2,141.
- **Tokenisation**: SentencePiece BPE, trained on the 200k subsampled sentences. Vocabulary size 8,000 per language.
- **Hardware**: NVIDIA Tesla T4 (15 GB VRAM), GCP `asia-east1`.
- **Optimizer**: Adam, lr $= 1 \times 10^{-3}$, ReduceLROnPlateau (patience=2, factor=0.5).
- **BLEU**: sacreBLEU corpus BLEU on BPE-detokenised output.

6 Training Results

6.1 Full Training Curve

Ep	Train	NMT	Cyc	Val	EN→DE	DE→EN	λ	s
1	13.5129	13.5129	14.4186	11.2766	0.79	0.94	0.00	561
2	12.3895	12.3895	13.6074	10.3556	1.46	1.61	0.00	532
3	11.9302	11.9302	13.5173	9.7761	2.05	2.56	0.00	523
4	11.6298	11.6298	13.5422	9.4027	3.00	3.57	0.00	524
5	11.4216	11.4216	13.5770	9.1186	3.17	4.74	0.00	521
6	11.9028	11.2686	12.6840	8.9393	3.53	4.49	0.05	523
7	11.7602	11.1385	12.4340	8.7374	4.07	5.87	0.05	522
8	11.6484	11.0335	12.2993	8.6012	4.16	6.22	0.05	522
9	11.5483	10.9389	12.1876	8.5136	4.11	6.66	0.05	526
10	11.4599	10.8549	12.0996	8.3821	4.24	6.71	0.05	525
11	11.9762	10.7920	11.8421	8.2916	4.75	7.01	0.10	522
12	11.8887	10.7273	11.6139	8.2286	4.74	6.85	0.10	520
13	11.8194	10.6739	11.4551	8.1692	4.90	7.21	0.10	520
14	11.7557	10.6252	11.3050	8.0587	4.95	7.72	0.10	522
15	11.6993	10.5822	11.1709	8.0503	5.24	7.75	0.10	525
16	11.6447	10.5408	11.0386	7.9873	5.17	7.95	0.10	520
17	11.5910	10.4998	10.9118	7.9320	5.28	8.13	0.10	522
18	11.5425	10.4633	10.7924	7.9029	5.19	7.89	0.10	521
19	11.5009	10.4340	10.6688	7.8330	5.30	8.31	0.10	521
20	11.4610	10.4062	10.5478	7.8163	5.52	8.70	0.10	521

Best checkpoint: **epoch 20**, val loss = **7.8163**. Val loss decreased every single epoch — 20 consecutive improvements with no oscillation.

6.2 Key Observations from Training Curve

Epochs 1–5 ($\lambda = 0.00$, translation only). NMT loss equals total train loss exactly (no cycle term). Val loss dropped from 11.28 to 9.12 — a 19.1% reduction in five epochs. BLEU on both directions climbed steadily: EN→DE reached 3.17 and DE→EN reached 4.74 by epoch 5. DE→EN consistently outperformed EN→DE from epoch 1 onward.

Epochs 6–10 ($\lambda = 0.05$, gentle inversion). Total train loss temporarily increased at epoch 6 (11.42 → 11.90) as the cycle consistency loss was introduced, then decreased monotonically. The NMT loss continued falling independently (11.27 → 10.85). Val loss continued improving to 8.38. DE→EN reached 6.71 by epoch 10.

Epochs 11–20 ($\lambda = 0.10$, full inversion constraint). The same pattern repeated at epoch 11: train loss bumped (11.46 → 11.98) then resumed decline. Val loss improved from 8.29 to 7.82. Both BLEU scores continued climbing: EN→DE reached 5.52 and DE→EN reached 8.70 on the validation set by epoch 20. The model had not plateaued by the final epoch, suggesting further improvement was available with additional training.

6.3 Final Test Results

Metric	Score
BLEU EN→DE (forward, uses $\mathbf{U}$)	4.10
BLEU DE→EN (backward, uses $\mathbf{U}^\top$)	6.44
Average BLEU	5.27
Best val loss	7.82

6.4 Cross-Experiment Comparison

Metric	v1	v2	v3	v4
EN→DE BLEU (test)	5.37	≈ 0	0.01	4.10
DE→EN BLEU (test)	3.47	≈ 0	0.00	6.44
Average BLEU (test)	4.42	≈ 0	0.01	5.27
Best val loss	17.08	14.18	12.61	7.82
Monotonic val loss	No	No	Yes	Yes
BLEU > 0 both dirs	Yes	No	No	Yes
DE→EN $>$ EN→DE	No	—	—	Yes

7 Translation Quality Analysis

7.1 Forward Path: EN→DE (uses U)

English Source	German Output
the president signed the agreement on monday .	das wort hat am montag .
scientists have discovered a new treatment for the disease .	wissenschaftlernen sich die krankheit fur die krankheitserklarung fur die krankheit .
the stock market fell sharply after the announcement .	die stockmarkt nach der ankunft des ersatzteil der nachweisungsausschusses nachkundigung .
thousands of people gathered in the city center to protest .	tausende menschen in der stadtstadt gebieten in der stadt center .
the team won the championship after a difficult season .	das teams hat sich am ende oktober nach einem schwierigen saison .
children need access to quality education in all regions .	kinder mussen zugang zu qualitat, bildung in allen regionen .
the company reported record profits in the third quarter .	die firma gegrundete konzertultat in der dritten viertel .
police arrested three suspects in connection with the attack .	polizei verurteilung der gemeinde in verbindung mit dem angriff zu den angriff auf den angriff .

Analysis. The forward path shows partial understanding. Content words are sometimes correctly identified ("kinder mussen zugang zu qualitat, bildung in allen regionen" is recognisable German for the children/education sentence). Recurring failure modes: (1) hallucination loops in longer sentences ("angriff auf den angriff"); (2) spurious BPE compound concatenations ("krankheitserklarung",

"stadtstadt"); (3) content drift where key verbs are replaced with unrelated words.

7.2 Backward Path: DE→EN (uses $\mathbf{U}^\top$)

German Source	English Output
der prasident hat das abkommen am montag unterzeichnet .	the president has the agreement on monday , .
wissenschaftler haben eine neue behandlung fur die krankheit entdeckt .	scientists have a new treatment for disease .
die aktienmarkte sind nach der ankundigung stark gefallen .	the shares markets are still available in the case of the attacks .
tausende von menschen versammelten sich im stadtzentrum , um zu protestieren .	thousands of people are interested in the city , resettlement to protest .
die mannschaft gewann die meisterschaft nach einer schwierigen saison .	the manhip of the masterplanner will be a difficult season .
kinder brauchen zugang zu qualitativ hochwertiger bildung in allen regionen .	children need access to qualitative education in all regions .
das unternehmen meldete rekordgewinne im dritten quartal .	the company is a key source of the third quarter of the third quarter of the third quarter quarter .
die polizei verhaftete drei verdachtige im zusammenhang mit dem angriff .	the police arrested three doubts in connection with the attacker .

Analysis. The backward path via $\mathbf{U}^\top$ produces clearly stronger output. Five of eight sentences are substantially correct: "scientists have a new treatment for disease", "children need access to qualitative education in all regions", "the president has the agreement on monday", and "the police arrested three doubts in connection with the attacker" all convey the correct meaning. Failure modes are milder — word-sense errors ("doubts" for "suspects", "qualitative" for "high-quality") rather than structural collapse.

7.3 Why DE→EN Outperforms EN→DE

The consistent superiority of the backward path (DE→EN BLEU 6.44 vs EN→DE BLEU 4.10) is a direct empirical finding of the Adigun-Kosko inversion framework. Three compounding factors explain it:

1. **Training data asymmetry.** WMT14 EN-DE is an English-dominant corpus; original documents are primarily English with professional German translations. German sentences are structurally more regular and translation-literal, making them easier to encode into invertible representations via $\mathbf{U}^\top$.

2. **German morphological explicitness.** German case suffixes, compound words, and rule-governed verb placement provide the encoder with consistent, decodable structure. These explicit markers produce hidden states that carry more recoverable information than English, which relies heavily on word order and function words.

3. **Inversion-specific supervision from cycle loss.** The cycle loss $\mathcal{L}_{\text{cyc,EN}}$ measures how well $\mathbf{U}^\top$ undoes what $\mathbf{U}$ did. The backward path receives gradient signal from both DE→EN

translation loss and both cycle consistency losses simultaneously, making $\mathbf{U}^\top$ better calibrated as an encoder-decoder than $\mathbf{U}$.

This asymmetry is an empirical prediction of the framework, not a flaw. It confirms that the B-BP inversion is working as theorised: the backward path is genuinely inverting the forward path's learned representation.

8 Discussion

8.1 Core B-BP Claim: Empirically Validated

Across all four experiments, the B-BP cell itself was never the source of failure. Every breakdown in v2 and v3 traced to components surrounding the cell — tokenisation, tied output projections, input feeding. The val loss trajectory confirms that internal representations improved consistently through all versions (32.51 → 17.08 in v1; 14.27 → 12.61 in v3; 11.28 → 7.82 in v4). The B-BP encoder was always learning. The breakdown was at the interface between the learned representation and the output vocabulary.

In v4, with that interface corrected (untied output projections), BLEU immediately became positive from epoch 1 and improved monotonically, confirming that the representations learned by the B-BP encoder were translation-relevant throughout.

8.2 The $\mathbf{U}^\top$ Inversion Result

The most significant finding is that DE→EN via $\mathbf{U}^\top$ achieves 6.44 BLEU on the test set, outperforming both the v1 baseline (3.47) and the forward path in v4 (4.10). A single recurrent matrix trained for forward translation can be transposed to produce a better translator in the reverse direction. This is the Adigun-Kosko inversion operating successfully in a practical NMT setting.

8.3 Remaining Limitations

1. **Absolute BLEU is modest.** 4–6 BLEU is low by production standards (state-of-the-art transformer models achieve 28–30 BLEU on WMT14 EN-DE). The constraint to a single recurrent matrix necessarily limits capacity. The goal of this work is to validate the B-BP inversion principle, not to maximise BLEU.
2. **Training not converged.** Val loss at epoch 20 was still decreasing. The model had not plateaued, suggesting 30 or more epochs would yield further improvement.
3. **Vocabulary size.** 8,000 BPE tokens is conservative for WMT14 news text. A larger vocabulary (16k–32k) would reduce OOV rate further and improve output fluency.

8.4 Next Steps

Change	Expected Effect	Rationale
Train for 30 epochs	+1–2 BLEU	Model not converged at ep 20
BPE vocab 8k → 16k	Improved fluency	Fewer OOV compounds
Orthogonality regulariser	Cleaner inversion	Pushes $\mathbf{U}$ toward $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$
Beam search ($k = 4$)	+1–2 BLEU	Greedy decoding suboptimal

9 Summary

Experiment v4 establishes the first clean B-BP NMT result in this series:

- EN→DE BLEU **4.10** and DE→EN BLEU **6.44** on the WMT14 test set, with average BLEU **5.27** surpassing the v1 baseline of 4.42.
- Val loss decreased for all 20 consecutive epochs (11.28 → 7.82), confirming stable and conflict-free training.
- The backward path via $\mathbf{U}^\top$ outperforms the forward path via $\mathbf{U}$ throughout training, providing empirical validation of the Adigun-Kosko inversion principle.
- Two root causes of prior failure were identified and fixed: untying output projections freed the embedding matrix from conflicting objectives; removing input feeding eliminated the decoder fixed-point collapse mechanism.